

Main Text — Tables and Figures

Racial and Ethnic Differences in Breast Cancer Incidence in the United States: A Systematic Review with Quantitative Synthesis

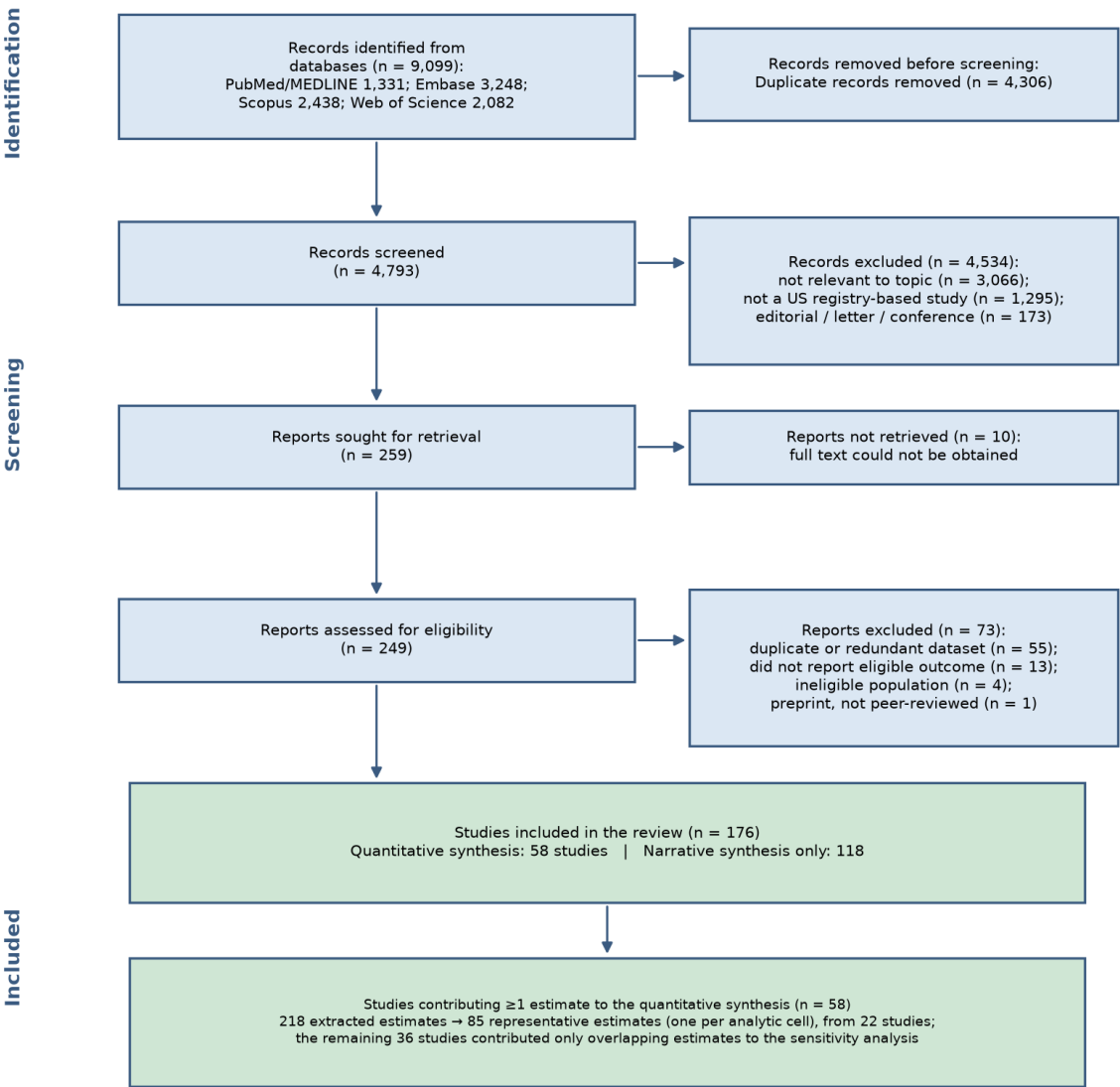
Table 1. Incidence rate ratios of invasive breast cancer among U.S. racial and ethnic groups relative to non-Hispanic White women, by analytic dimension

Group	Effect	Estimate [95% CI]	Representative study	Registry family	RoB
Overall invasive breast cancer					
Hispanic	IRR	0.718 [0.707, 0.731] ‡	Ellington 2022[25] (2018)	USCS(NPCR+SEER ~99%)	Low
Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	IRR	0.769 [0.752, 0.788] ‡	Ellington 2022[25] (2018)	USCS(NPCR+SEER ~99%)	Low
American Indian and Alaska Native (AI/AN)	IRR	0.87 (point est.)	Melkonian 2019[53] (2010-2015)	IHS-PRCDA	Moderate
non-Hispanic Black (NHB)	IRR	0.933 [0.920, 0.946] ‡	Ellington 2022[25] (2018)	USCS(NPCR+SEER ~99%)	Low
Asian American subgroups					
Hmong	IRR	0.164 [0.120, 0.224] ‡	Mills 2005[6] (1988-2000)	California-CCR	Low
Cambodian	IRR	0.264 [0.204, 0.341] ‡	Kem 2007[55] (1998-2002)	California-CCR	Moderate
Laotian/Kampuchean	IRR	0.388 [0.350, 0.430] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Vietnamese	IRR	0.586 [0.566, 0.608] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Korean	IRR	0.677 [0.654, 0.701] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Chinese	IRR	0.760 [0.745, 0.775] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Filipina	IRR	0.857 [0.839, 0.874] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Asian Indian/Pakistani	IRR	0.958 [0.937, 0.979] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Japanese	IRR	1.045 [1.009, 1.083] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Asian American (aggregate)	IRR	0.792 [0.784, 0.800] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Native Hawaiian and Pacific Islander (NHPI) subgroups					
Guamanian/Chamorro/Samoan	IRR	0.723 [0.641, 0.817] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Native Hawaiian	IRR	1.270 [1.213, 1.330] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Native Hawaiian and Pacific Islander (NHPI)	IRR	1.212 [1.168, 1.258] ‡	Gomez 2026[31] (2018-2022)	SEER-national	Moderate
Hispanic/Latina by country of origin					
Mexican	IRR	0.512 [0.382, 0.686] ‡	Pinheiro 2009[8] (1999-2001)	State: Florida	Low
New Latino	IRR	0.697 [0.629, 0.771] ‡	Pinheiro 2009[8] (1999-2001)	State: Florida	Low
Cuban	IRR	0.769 [0.688, 0.860] ‡	Pinheiro 2009[8] (1999-2001)	State: Florida	Low
Puerto Rican	IRR	0.833 [0.738, 0.939] ‡	Pinheiro 2009[8] (1999-2001)	State: Florida	Low
American Indian and Alaska Native (AI/AN) by region					
AI/AN (Navajo)	IRR	0.49 [0.44, 0.55]	Yazzie 2025[48] (2014-2018)	IHS-PRCDA	Low
AI/AN (Southwest)	IRR	0.57 (point est.)	Melkonian 2019[53] (2010-2015)	IHS-PRCDA	Moderate
AI/AN (East)	IRR	0.61 (point est.)	Melkonian 2019[53] (2010-2015)	IHS-PRCDA	Moderate
AI/AN (Pacific Coast)	IRR	0.93 (point est.)	Melkonian 2019[53] (2010-2015)	IHS-PRCDA	Moderate
AI/AN (Northern Plains)	IRR	1.05 (point est.)	Melkonian 2019[53] (2010-2015)	IHS-PRCDA	Moderate
Alaska Native	IRR	1.09 [0.99, 1.21] †	Nash 2019[11] (2009-2014)	Alaska Native Tumor Registry (IHS-linked)	Low
AI/AN (Southern Plains)	IRR	1.33 [1.26, 1.41]	Melkonian 2021[58] (2012-2016)	IHS-PRCDA	Low
Middle Eastern					
Middle Eastern	IRR	0.86 [0.84, 0.88]	Nasseri 2009[46] (1988-2004)	California-CCR	Moderate

Group	Effect	Estimate [95% CI]	Representative study	Registry family	RoB
Triple-negative breast cancer					
Chinese	IRR	0.53 [0.31, 0.93] † ‡	Loo 2019[24] (2010-2013)	Hawaii Tumor Registry (SEER)	Low
Asian American (aggregate)	IRR	0.61 [0.58, 0.65]	Gomez 2017[35] (2009-2013)	California-CCR	Low
Asian American, Native Hawaiian, and Pacific Islander (AANHPI)	IRR	0.700 [0.680, 0.720]	Sung 2023[23] (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Filipina	IRR	0.84 [0.60, 1.18] † ‡	Loo 2019[24] (2010-2013)	Hawaii Tumor Registry (SEER)	Low
American Indian and Alaska Native (AI/AN)	IRR	0.860 [0.800, 0.930]	Sung 2023[23] (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Hispanic	IRR	0.860 [0.840, 0.880]	Sung 2023[23] (2015-2019)	USCS(NPCR+SEER ~99%)	Low
Native Hawaiian	IRR	0.86 [0.60, 1.22] † ‡	Loo 2019[24] (2010-2013)	Hawaii Tumor Registry (SEER)	Low
Japanese	IRR	1.08 [0.79, 1.46] † ‡	Loo 2019[24] (2010-2013)	Hawaii Tumor Registry (SEER)	Low
non-Hispanic Black (NHB)	IRR	1.950 [1.930, 1.980]	Sung 2023[23] (2015-2019)	USCS(NPCR+SEER ~99%)	Low

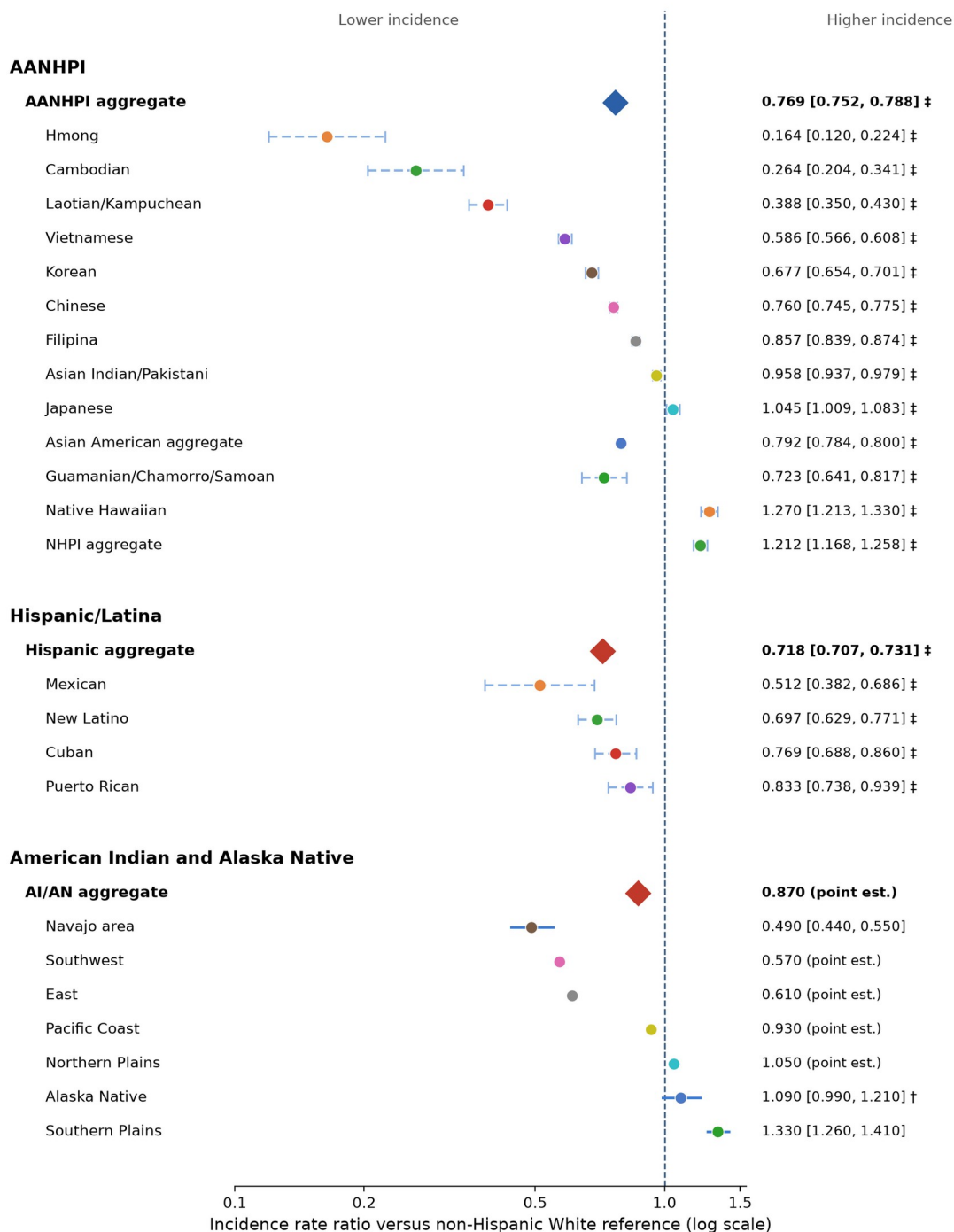
Note. Each cell shows one representative population-based estimate — the most recent, broadest-coverage registry estimate per analytic cell — not a pooled estimate; the selection rule and its robustness are given in the Methods and Supplementary Table 6. Each representative study is followed by its number in the reference list. The effect measure is the incidence rate ratio (IRR) unless marked as a standardized incidence ratio (SIR); comparisons are versus non-Hispanic White (NHW) women. † the study's reference was an unstratified White group (not stratified by Hispanic origin), examined in the NHW-comparator sensitivity analysis (Supplementary Table 6c). ‡ the 95% confidence interval was computed by the reviewers from published rates rather than reported in the source. NHB, non-Hispanic Black; AANHPI, Asian American, Native Hawaiian, and Pacific Islander (NHPI, the Pacific Islander subset, shown separately); RoB, risk-of-bias rating of the representative study. Full per-estimate detail is in the Supplementary Materials.

Figure 1. PRISMA 2020 flow diagram



PRISMA 2020 flow diagram. Identification via databases only; no registers, citation searching, or grey-literature sources were used. Single-reviewer screening with AI assistance. Search date 7 August 2026.

Figure 2. Aggregate-to-disaggregated heterogeneity in breast cancer incidence (representative IRR vs non-Hispanic White, 95% CI): the Asian American, Native Hawaiian and Pacific Islander (AANHPI), Hispanic/Latina, and American Indian and Alaska Native (AI/AN) aggregates, each shown with its subgroups



Note. Each point is the representative estimate for one analytic cell, drawn from a separate study; an aggregate and its subgroups differ in registry, region, diagnosis period, and standard population. Diamonds mark aggregate groups, circles subgroups. A solid bar is a 95% CI reported in the source; a dashed bar marked ‡ is one computed for this review; a point without a bar had no interval in its source. † reference was an unstratified White group, not NHW. Values match Table 1.

Figure 3. Incidence rate ratio by racial or ethnic group and analytic dimension (representative estimate vs non-Hispanic White; 1.0 = the NHW rate)

